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ESSnet Big Data II

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https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata
https://ec.europa.eu/eurostat/cros/content/essnetbigdata_en

Workpackage G

Financial transaction data

Deliverable 3: Closing the selection of case studies

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Workpackage country contact:

Valentin Chavdarov (BG, Bulgaria)
Nathalie Wiersma and Christian Janz (DE, Germany)
Alessandra Righi (IT, Italy)
Johan Fosen (NO, Norway)
Pedro Campos (PT, Portugal)
Snežana Vrhovac (SI, Slovenia)

Contact in the cooperation with Banca d'Italia: Giuseppe Bruno

Workpackage Leader:

Johan Fosen (SSB, Norway)
johan.fosen@ssb.no
telephone: +47 990 17 864
mobile phone: +47 990 17 864

1. Background

The Essnet Big Data II work package "WPG Financial transactions" contains two steps according to the grant agreement:

1) Describing the financial transaction data infrastructure and metadata in each of the countries as well as identifying promising official statistics that can benefit from financial transaction data. The interim findings of the first part are described in the WPG Deliverable 1[1].
2) Performing empirical studies among promising official statistics. The plan for the case studies were described in WPG Deliverable 2[2].

Deliverable 3 is part of step 2 and provides an overview of the case studies that will be performed as a part of WPG, thus those case studies will be completed within the time frame of the WPG project. However, further case studies might be included after discussion with Eurostat, if there is sufficient amount of resources and calendar time left for completing such case studies.

2. The list of case studies performed within WPG

The below list consists of the case studies that have either started or will start soon with a high probability. The third and fourth case studies (Section 2.3 and Section 2.4) are confirmed, whereas for case studies 2.1 and 2.2 the use of NEXI information for the purposes of the ESSnet Project is subject to approval. A formal consent request was submitted by the Bank of Italy to managers of the private Card processor based on the information contained in this Deliverable 3.

2.1 Italian case study 1

- The Italian case study 1 is about using financial transaction data for macroeconomic forecasts.
- The financial transaction data used are:
    - From the NEXI Card processor: time series of total international credit/debit card transactions (daily or monthly recorded) by channel (e-commerce / proximity) and by main MCC aggregates (express in value and in number).
    - Bank of Italy BI-COMP database: time series of total transactions and total domestic debit card transactions (proximity channel).
- The aim of this case study is to use the aggregated time series of NEXI and Bank of Italy financial transaction data to forecast retail trade or private consumption.

2.2 Italian case study 2

- The Italian case study 2 is about the integration of the amount of the expenditure of the Italians for (domestic and outbound) tourism and of the foreigners traveling in Italy (inbound tourism) in the Tourism Satellite Account.
- The financial transaction data used are from NEXI card processor source: 2015 and 2017 annual amount (in value and in number) relating to:
    - total transactions of Italian credit/debit card (proximity channel) for expenditure made abroad by aggregated categories of MCC related to the tourism (e.g. Hotels and restaurants, Travels and transports, or possibly more granular breakdowns)

[1] https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/c/c9/WPG_Deliverable_G1_Interim_findings_for_financial_transaction_data_and_sharing_economy_2020_01_07.pdf
[2] https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/f/f3/WPG_Deliverable_G2_Conclusions_on_studies%2C_and_plan_for_year_two_201_11_08.pdf

    - total transactions of Italian credit/debit card (proximity channel) for expenditure made in Italy by aggregated categories of MCC related to the tourism (e.g., Hotels and restaurants, Travels and transports, or possibly more granular breakdowns) by Italian region where the payment is made
    - total transactions of non-Italian credit/debit cards (proximity channel) for expenditure made in Italy by tourist categories by Italian region where the payment is made.
- The aim of this case study is to test the feasibility of the integration of these data in the construction of the Tourism Satellite Account for the years 2015 and 2017. The amounts of expenditure already calculated by the 2015 Satellite account will be compared to the annual amounts deriving from the FTD.

2.3 Norwegian case study

- The Norwegian case study is about the retail sales index statistics.
- The financial transaction data used is a dataset of national payment system debit card transactions at point-of-sales terminals for one day in 2016. These data give the total sales from this card scheme for each local business unit, which is the unit of interest in this case study.
- The purpose of the case study is to see to what extent and how the financial transaction data could improve today's estimation in this official statistics whose current data collection is a survey providing Statistics Norway with the total retail sales number monthly on local business unit level.

2.4 Portuguese case study

- The Portuguese case study is about accommodation statistics.
- The financial transaction data used is a data set received monthly from the Tax authorities and covers all transactions both at point of sales terminals as well as on Internet, paid by both cash, debit cards and credit cards. The unit of the data is the combination of payer and payee. For each unit, the data set contains the total transaction amount between the payer and payee during a month, thus corresponding to a single transaction only in the case where only one transaction has occurred between the payer and payee during the month.
- The purpose of the case study is to investigate accommodation due to sharing economy, which is not included in the accommodation statistics today.